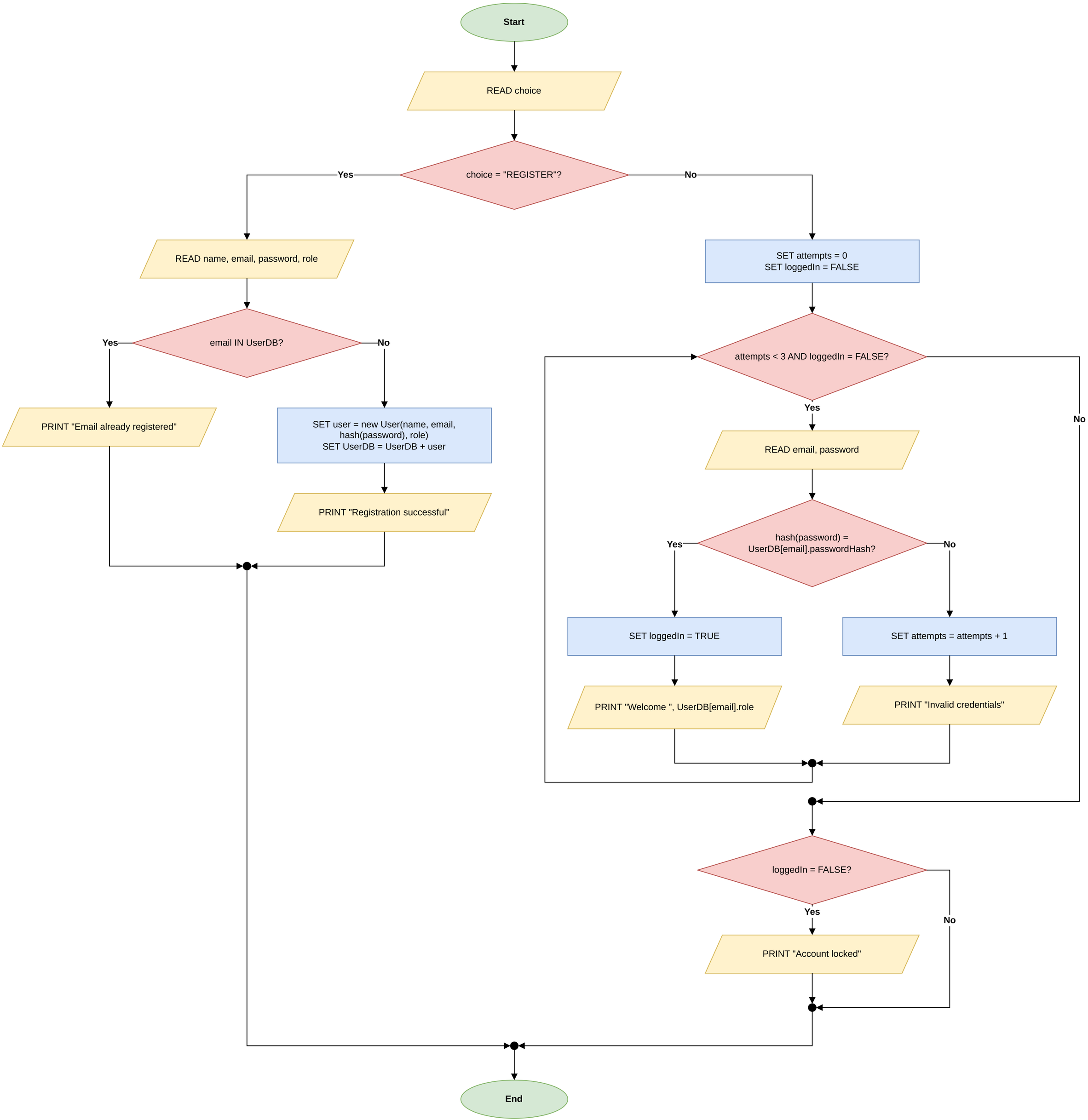
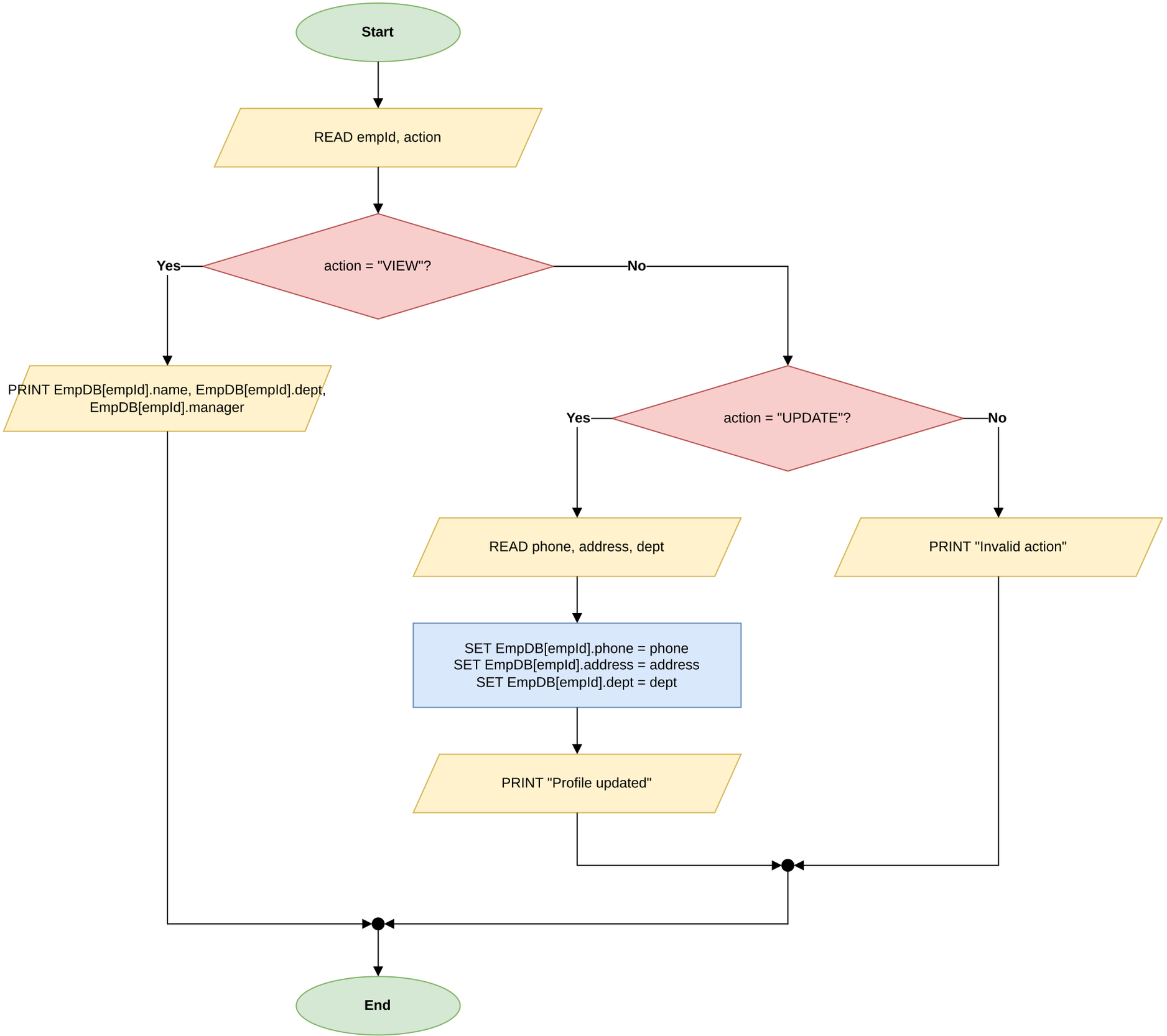
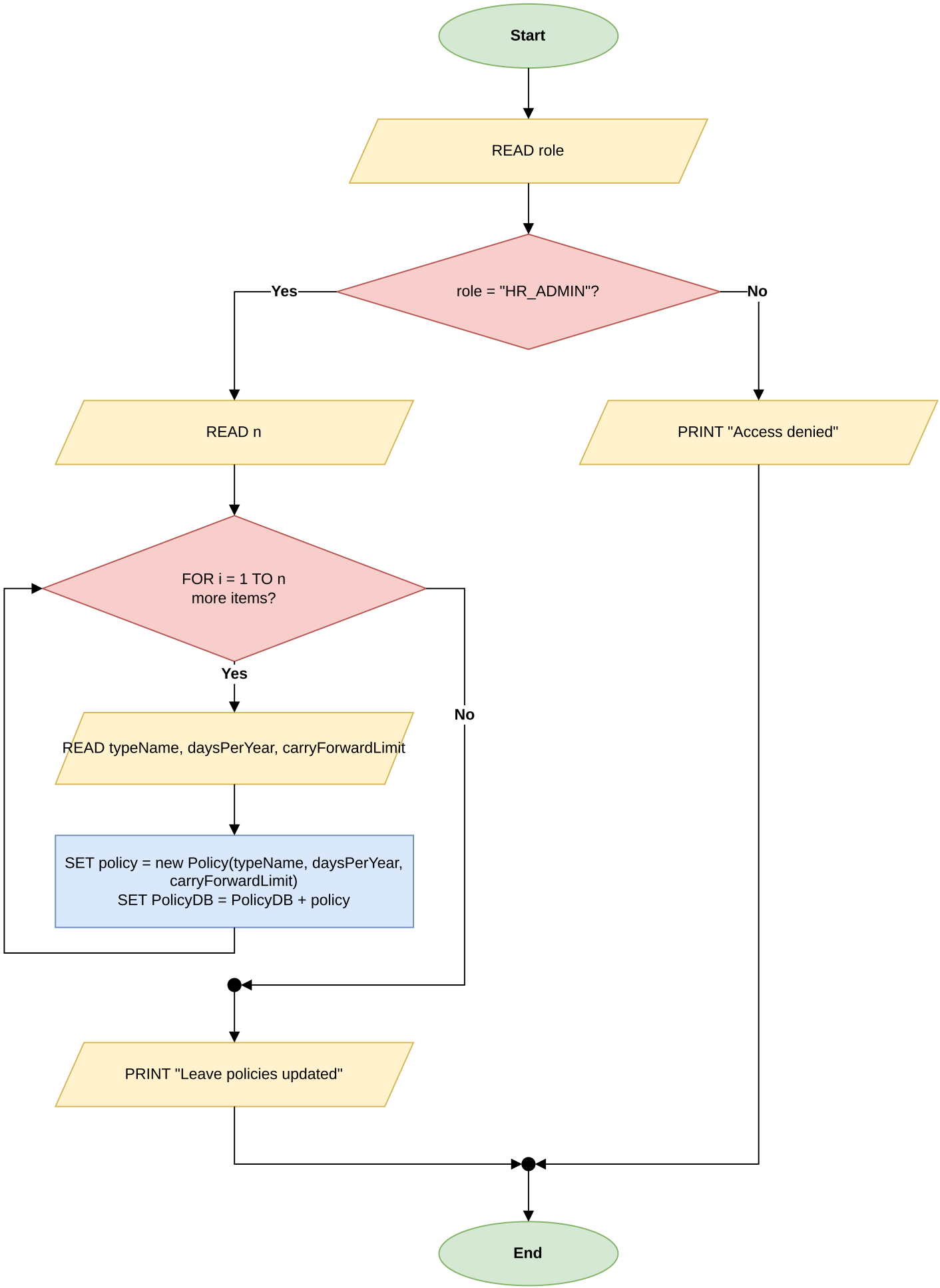
Module 1. User Registration & Login – Flowchart



Module 2. Employee Profile Management – Flowchart



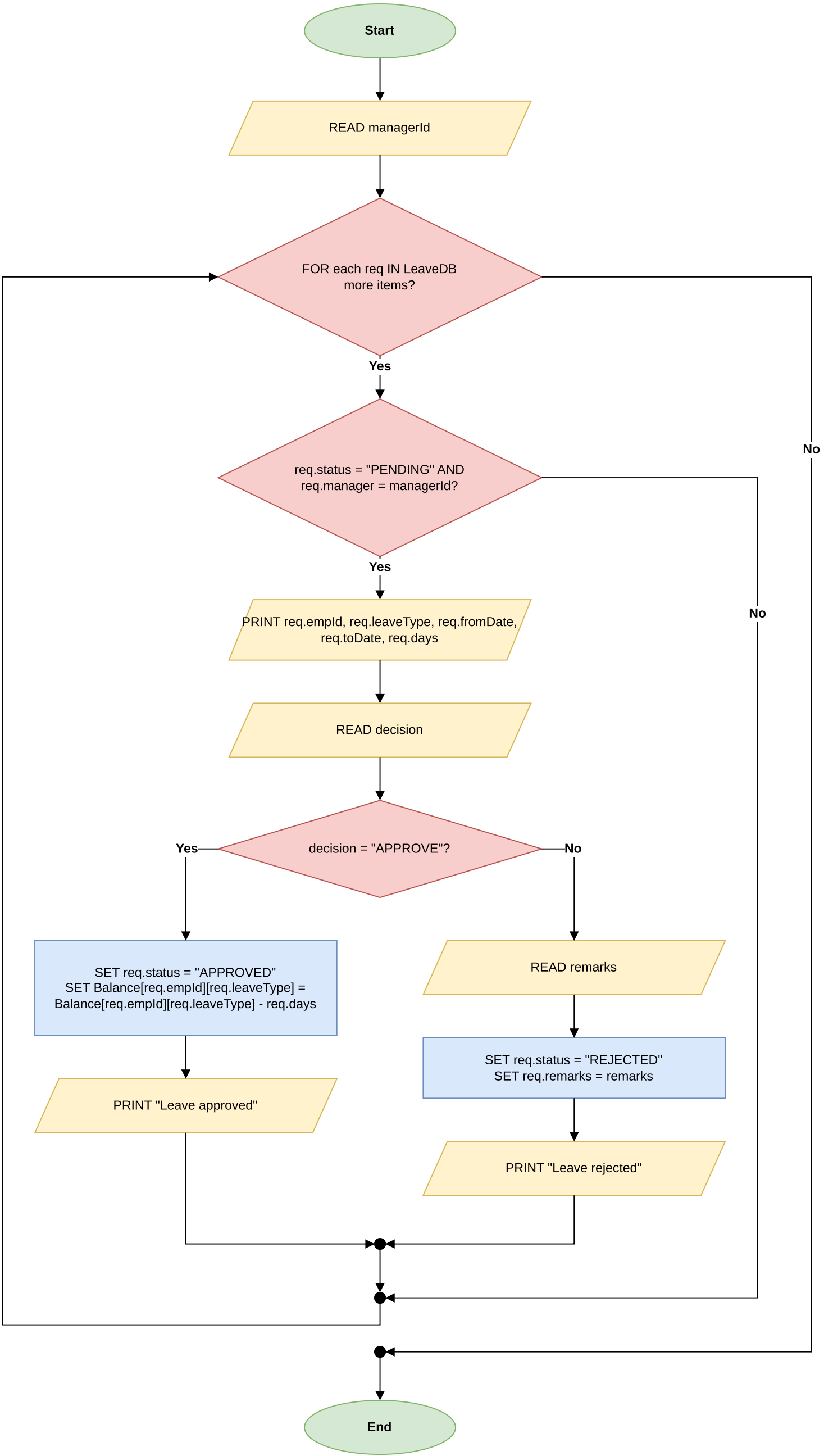
Module 3. Leave Type & Policy Management – Flowchart



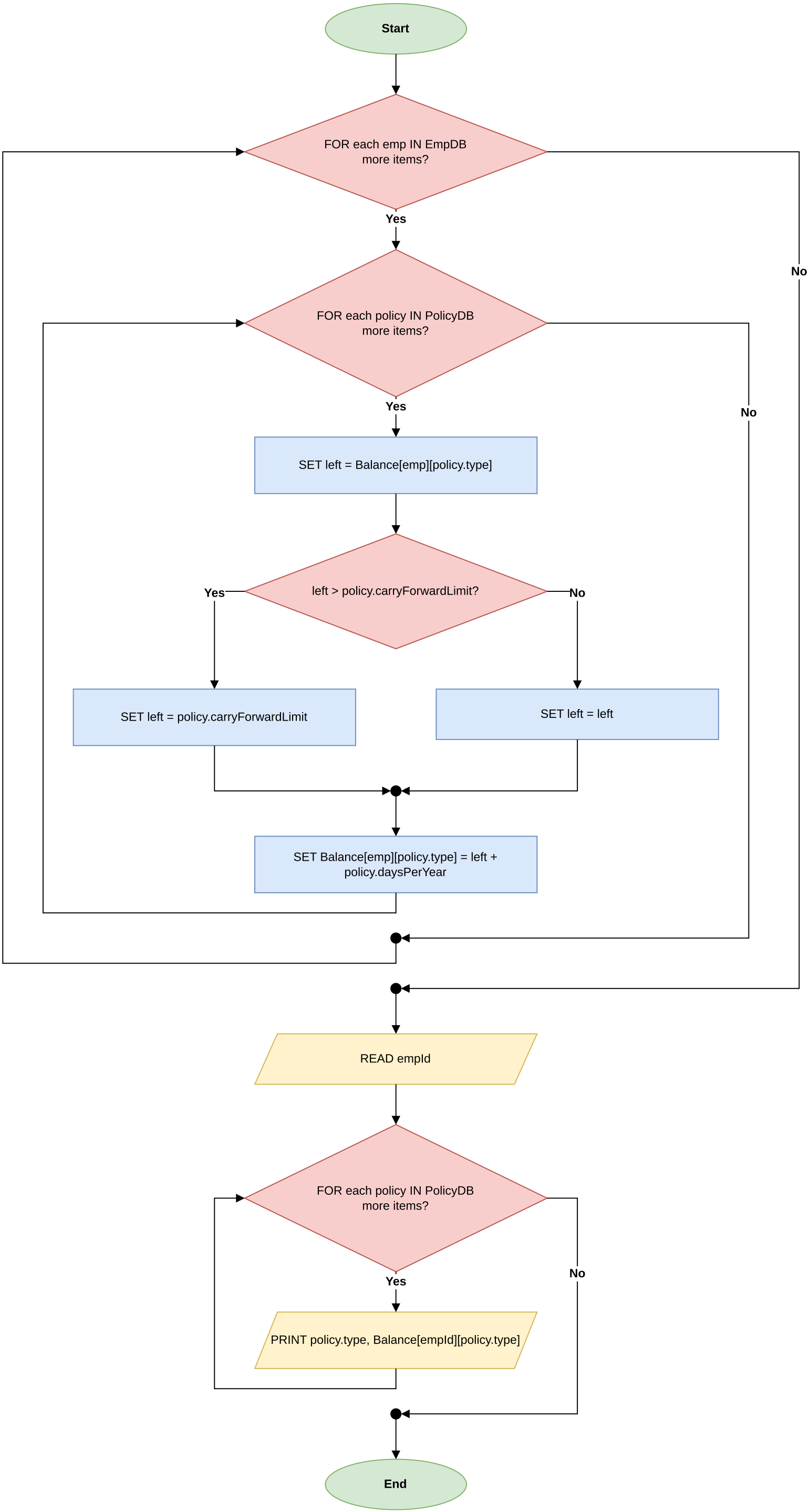
Module 4. Leave Application – Flowchart



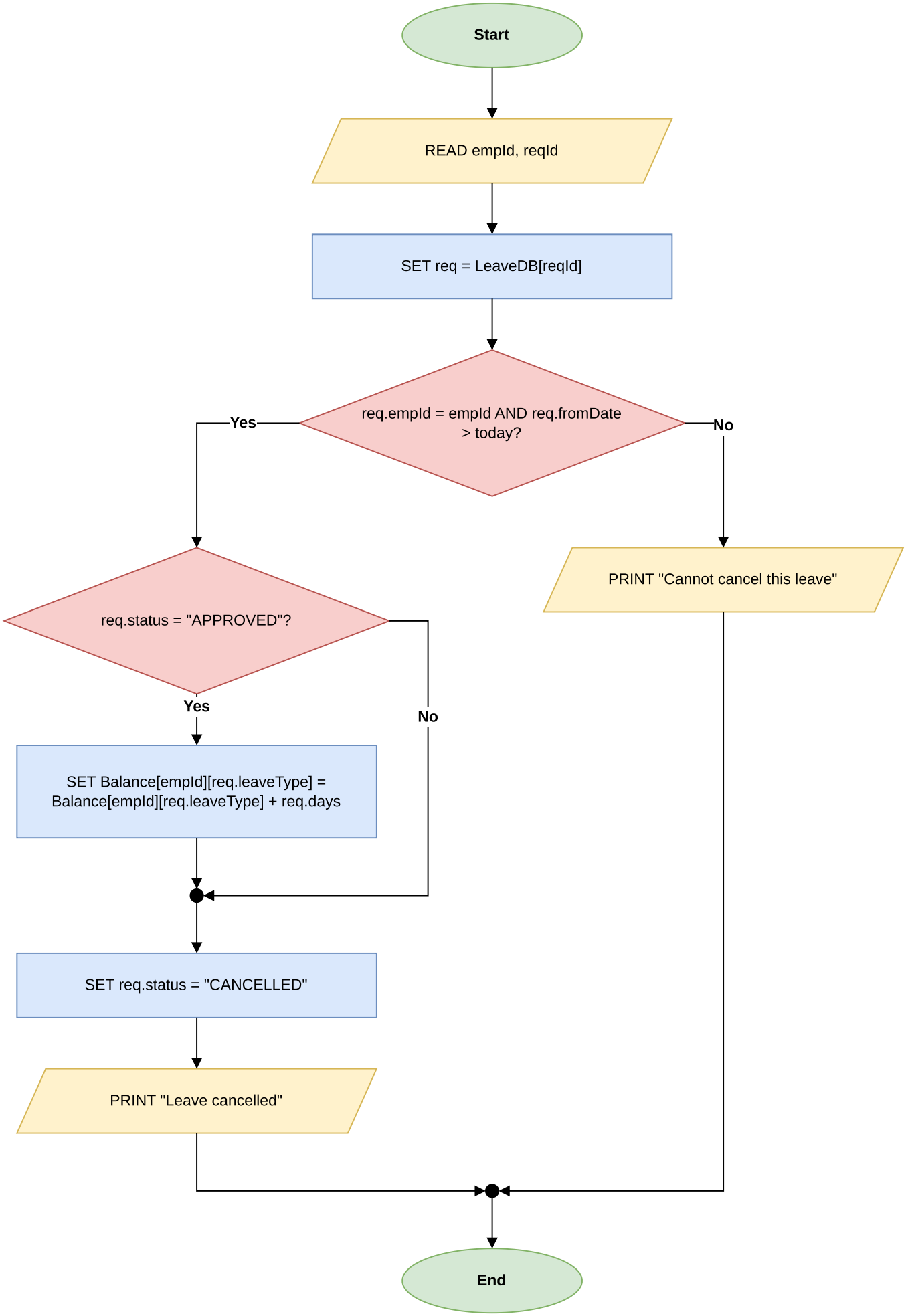
Module 5. Leave Approval / Rejection – Flowchart



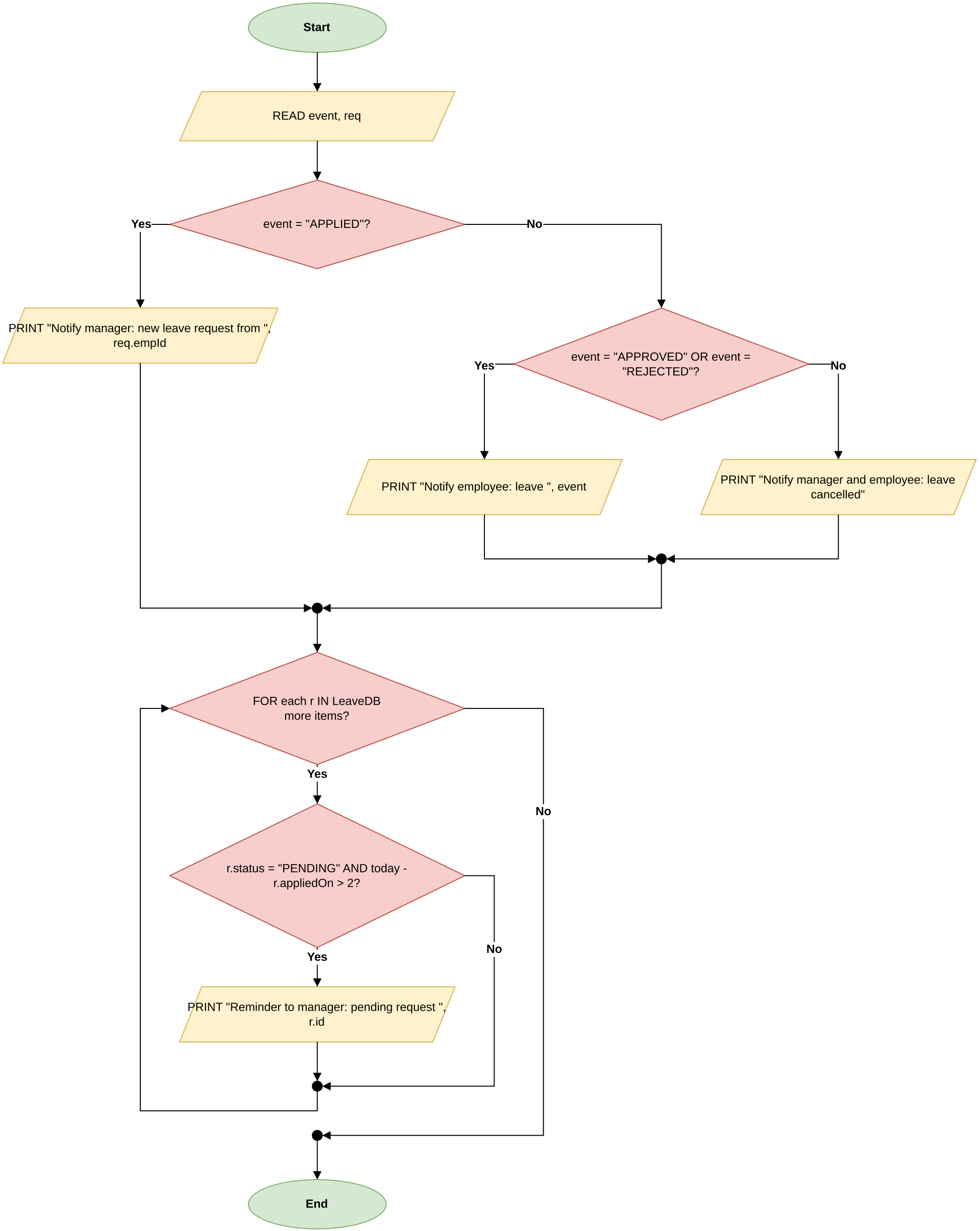
Module 6. Leave Balance Calculation – Flowchart



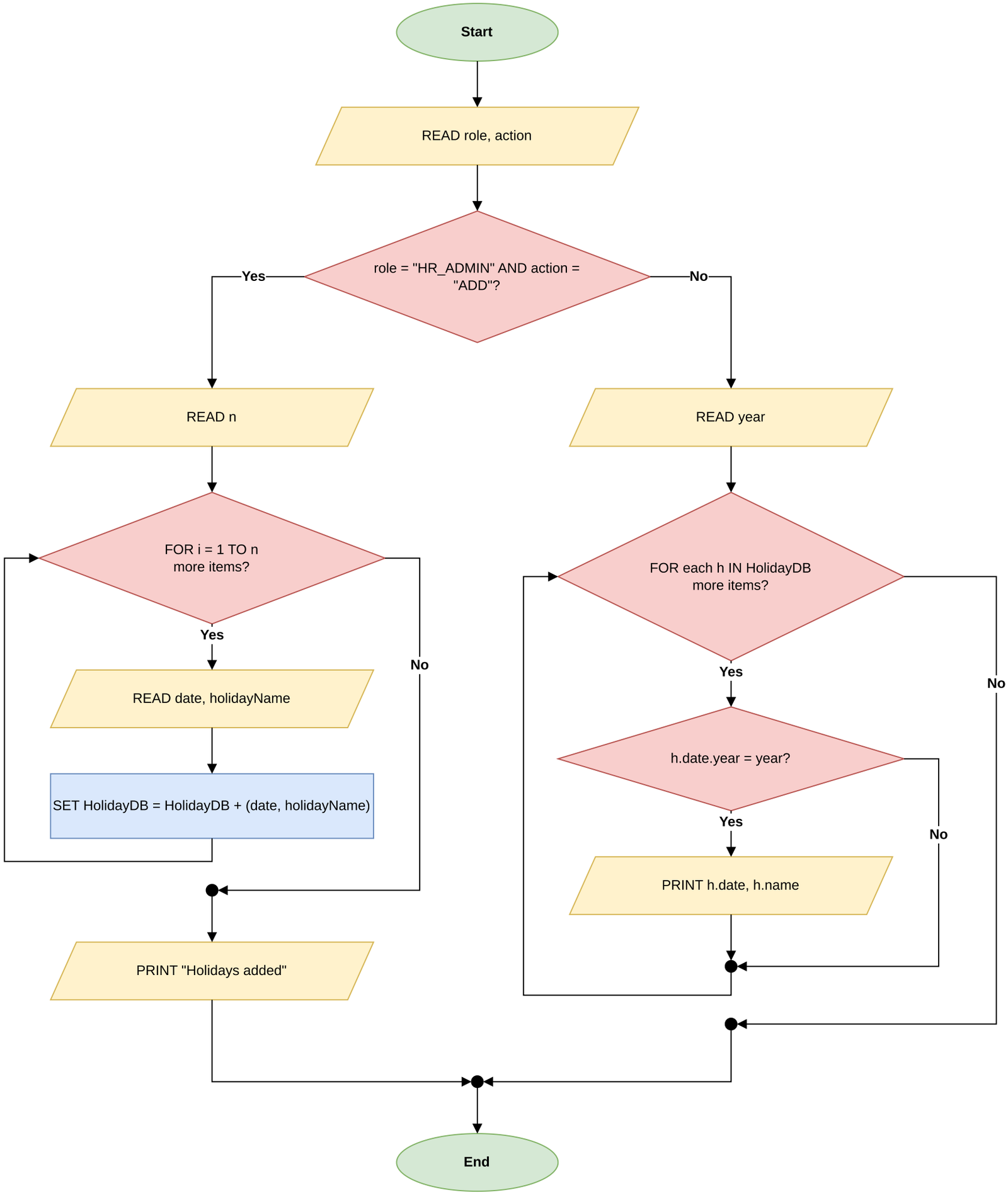
Module 7. Leave Cancellation – Flowchart



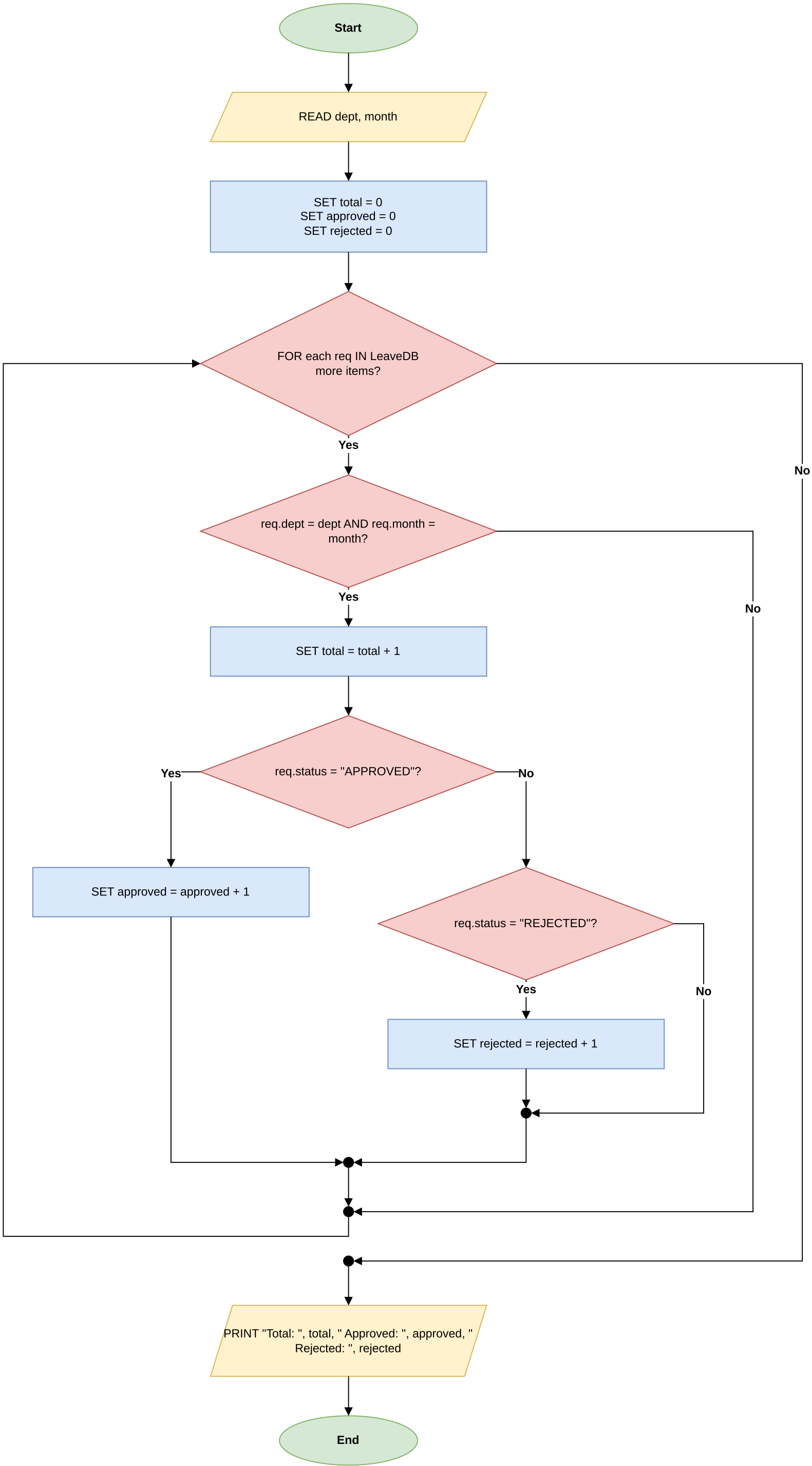
Module 8. Notification – Flowchart



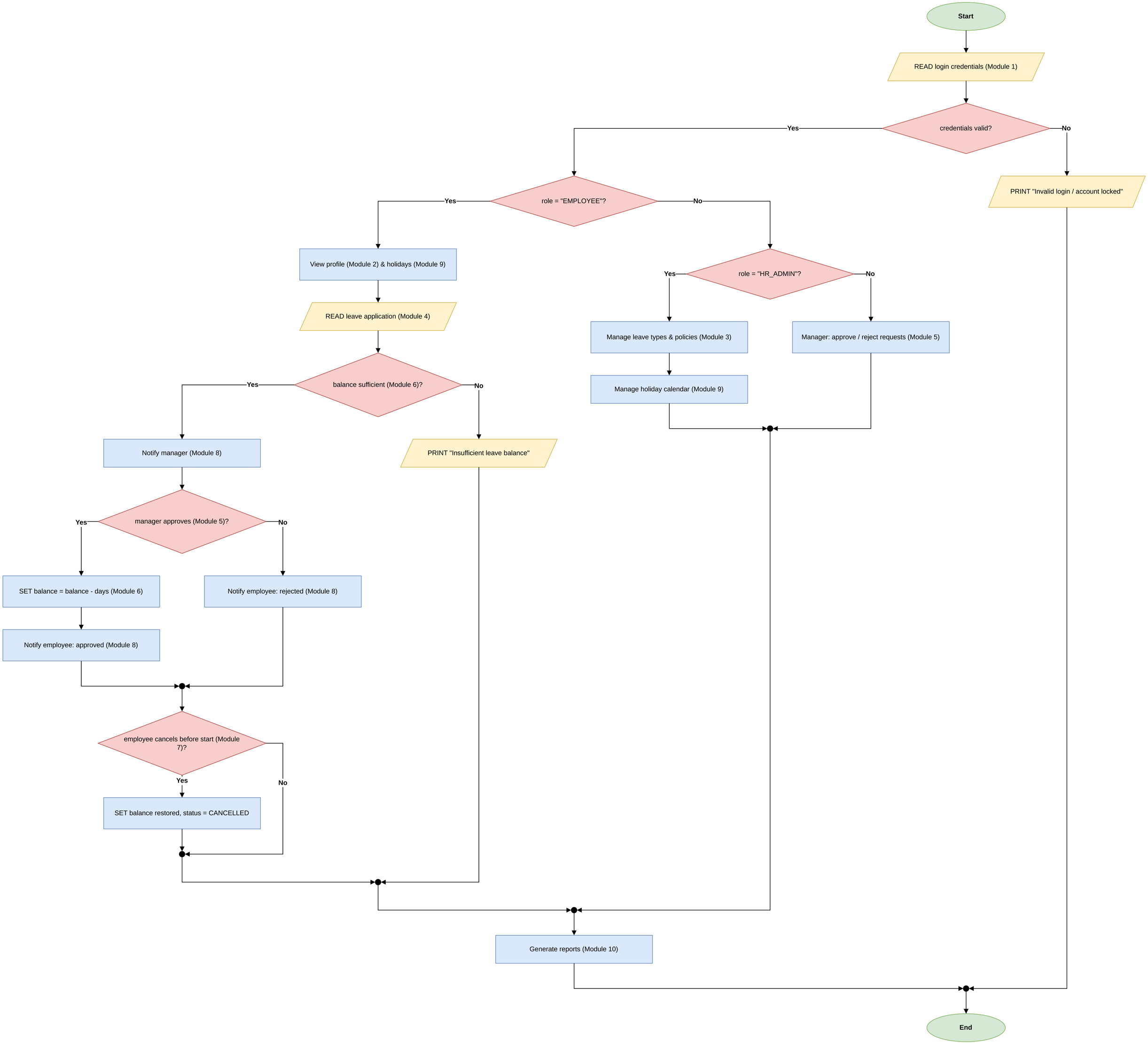
Module 9. Holiday Calendar – Flowchart

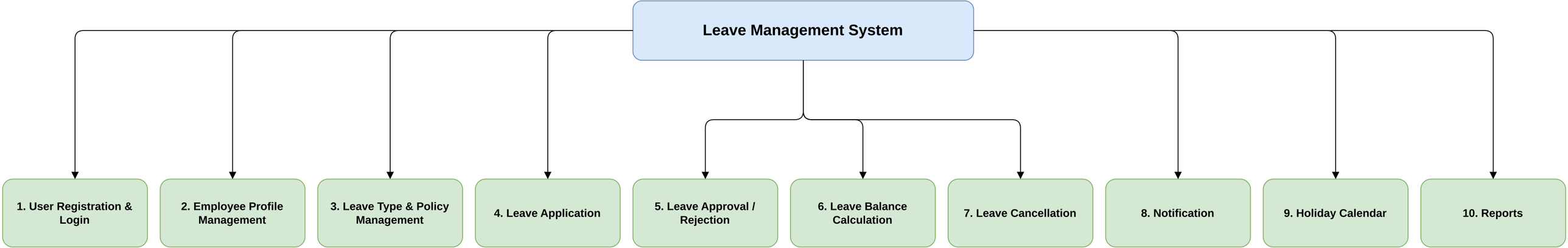


Module 10. Reports – Flowchart



Leave Management System – System Flowchart / Sequence Flow




Leave Management System
1. User Registration & Login
2. Employee Profile Management
3. Leave Type & Policy Management
4. Leave Application
5. Leave Approval / Rejection
6. Leave Balance Calculation
7. Leave Cancellation
8. Notification
9. Holiday Calendar
10. Reports

Module 1. User Registration & Login

```
BEGIN
READ choice
IF choice = "REGISTER" THEN
    READ name, email, password, role
    IF email IN UserDB THEN
        PRINT "Email already registered"
    ELSE
        SET user = new User(name, email, hash(password), role)
        SET UserDB = UserDB + user
        PRINT "Registration successful"
    END IF
ELSE
    SET attempts = 0
    SET loggedIn = FALSE
    WHILE attempts < 3 AND loggedIn = FALSE DO
        READ email, password
        IF hash(password) = UserDB[email].passwordHash THEN
            SET loggedIn = TRUE
            PRINT "Welcome ", UserDB[email].role
        ELSE
            SET attempts = attempts + 1
            PRINT "Invalid credentials"
        END IF
    END WHILE
    IF loggedIn = FALSE THEN
        PRINT "Account locked"
    END IF
END IF
END
```

Module 2. Employee Profile Management

```
BEGIN
READ empId, action
IF action = "VIEW" THEN
    PRINT EmpDB[empId].name, EmpDB[empId].dept, EmpDB[empId].manager
ELSE
    IF action = "UPDATE" THEN
        READ phone, address, dept
        SET EmpDB[empId].phone = phone
        SET EmpDB[empId].address = address
        SET EmpDB[empId].dept = dept
        PRINT "Profile updated"
    ELSE
        PRINT "Invalid action"
    END IF
END IF
END
```

Module 3. Leave Type & Policy Management

```
BEGIN
READ role
IF role = "HR_ADMIN" THEN
    READ n
    FOR i = 1 TO n DO
        READ typeName, daysPerYear, carryForwardLimit
        SET policy = new Policy(typeName, daysPerYear, carryForwardLimit)
        SET PolicyDB = PolicyDB + policy
    END FOR
    PRINT "Leave policies updated"
ELSE
    PRINT "Access denied"
END IF
END
```

Module 4. Leave Application

```
BEGIN
READ empId, leaveType, fromDate, toDate, reason
SET days = 0
SET d = fromDate
WHILE d <= toDate DO
    IF d IN HolidayDB OR d is weekend THEN
        SET days = days
    ELSE
        SET days = days + 1
    END IF
    SET d = d + 1
END WHILE
IF days <= Balance[empId][leaveType] THEN
    SET req = new Request(empId, leaveType, fromDate, toDate, days, reason)
    SET req.status = "PENDING"
    SET LeaveDB = LeaveDB + req
    PRINT "Leave applied for ", days, " days"
ELSE
    PRINT "Insufficient leave balance"
END IF
END
```

Module 5. Leave Approval / Rejection

```
BEGIN
READ managerId
FOR each req IN LeaveDB DO
    IF req.status = "PENDING" AND req.manager = managerId THEN
        PRINT req.empId, req.leaveType, req.fromDate, req.toDate, req.days
        READ decision
        IF decision = "APPROVE" THEN
            SET req.status = "APPROVED"
            SET Balance[req.empId][req.leaveType] = Balance[req.empId][req.leaveType] - req.days
            PRINT "Leave approved"
        ELSE
            READ remarks
            SET req.status = "REJECTED"
            SET req.remarks = remarks
            PRINT "Leave rejected"
        END IF
    END IF
END FOR
END
```

Module 6. Leave Balance Calculation

```
BEGIN
FOR each emp IN EmpDB DO
    FOR each policy IN PolicyDB DO
        SET left = Balance[emp][policy.type]
        IF left > policy.carryForwardLimit THEN
            SET left = policy.carryForwardLimit
        ELSE
            SET left = left
        END IF
        SET Balance[emp][policy.type] = left + policy.daysPerYear
    END FOR
END FOR
READ empId
FOR each policy IN PolicyDB DO
    PRINT policy.type, Balance[empId][policy.type]
END FOR
END
```

Module 7. Leave Cancellation

```
BEGIN
READ empId, reqId
SET req = LeaveDB[reqId]
IF req.empId = empId AND req.fromDate > today THEN
    IF req.status = "APPROVED" THEN
        SET Balance[empId][req.leaveType] = Balance[empId][req.leaveType] + req.days
    END IF
    SET req.status = "CANCELLED"
    PRINT "Leave cancelled"
ELSE
    PRINT "Cannot cancel this leave"
END IF
END
```

Module 8. Notification

```
BEGIN
READ event, req
IF event = "APPLIED" THEN
    PRINT "Notify manager: new leave request from ", req.empId
ELSE
    IF event = "APPROVED" OR event = "REJECTED" THEN
        PRINT "Notify employee: leave ", event
    ELSE
        PRINT "Notify manager and employee: leave cancelled"
    END IF
END IF
FOR each r IN LeaveDB DO
    IF r.status = "PENDING" AND today - r.appliedOn > 2 THEN
        PRINT "Reminder to manager: pending request ", r.id
    END IF
END FOR
END
```

Module 9. Holiday Calendar

```
BEGIN
READ role, action
IF role = "HR_ADMIN" AND action = "ADD" THEN
    READ n
    FOR i = 1 TO n DO
        READ date, holidayName
        SET HolidayDB = HolidayDB + (date, holidayName)
    END FOR
    PRINT "Holidays added"
ELSE
    READ year
    FOR each h IN HolidayDB DO
        IF h.date.year = year THEN
            PRINT h.date, h.name
        END IF
    END FOR
END IF
END
```

Module 10. Reports

```
BEGIN
READ dept, month
SET total = 0
SET approved = 0
SET rejected = 0
FOR each req IN LeaveDB DO
    IF req.dept = dept AND req.month = month THEN
        SET total = total + 1
        IF req.status = "APPROVED" THEN
            SET approved = approved + 1
        ELSE
            IF req.status = "REJECTED" THEN
                SET rejected = rejected + 1
            END IF
        END IF
    END IF
END FOR
PRINT "Total: ", total, " Approved: ", approved, " Rejected: ", rejected
END
```